Building Analytical Dashboard

Project Synopsis Report

Date: May 04, 2026

Project Type: BCA Final Year Project

1. Introduction

This project is an interactive analytics dashboard developed using Streamlit, Pandas, NumPy, and Plotly. It analyzes socio-economic indicators from the Gapminder dataset to help users understand global development patterns between 1952 and 2007. The dashboard enables dynamic exploration of country and continent-level trends through filtering, KPI cards, charts, and map-based visualizations.

2. Problem Statement

Large socio-economic datasets are difficult to interpret in tabular form. Decision-makers and learners need a simple and interactive system that transforms historical data into meaningful visual insights. This project solves that problem by offering a user-friendly dashboard for exploring population, life expectancy, and GDP patterns across different regions and years.

3. Objectives

- Build an end-to-end analytics dashboard using real-world data.
- Provide filtering by year, continent, and country for focused analysis.
- Present key metrics (population, life expectancy, GDP per capita) in summary form.
- Visualize relationships and trends using interactive charts and maps.
- Allow users to export filtered results for reporting and further analysis.

4. Dataset Description

The dashboard uses the Gapminder dataset provided by Plotly (px.data.gapminder()). It includes country-level records with fields such as country, continent, year, population, life expectancy, GDP per capita, and ISO country code. The application also derives a new metric: Total GDP = GDP per Capita x Population.

5. Technology Stack

Technology	Purpose
Python	Core programming language

Streamlit	Web dashboard framework
Pandas	Data handling and transformation
NumPy	Weighted calculations and numeric operations
Plotly	Interactive charts and map visualizations

6. System Workflow

1. Load Gapminder data and compute derived total GDP.
2. Accept user filters from sidebar controls.
3. Apply continent and country filters, then isolate selected year.
4. Compute aggregate KPIs using weighted and total metrics.
5. Render charts, map, and top-country comparisons.
6. Display filtered table and provide CSV download.

7. Functional Modules

The application is organized into the following functional modules:

- Data Loading Module: Uses cached loading for faster repeated usage.
- Filter Module: Sidebar controls for year, continent, and optional country selection.
- KPI Module: Computes countries count, total population, weighted life expectancy, and weighted GDP per capita.
- Visualization Module: Includes scatter, bar, line, area, choropleth map, and top-10 GDP charts.
- Data Export Module: Exports filtered table in CSV format for reuse.

8. Key Visual Analytics

GDP vs Life Expectancy Bubble Chart: Highlights economic-health relationship with bubble size as population.
Population by Continent Bar Chart: Compares continental population shares.
Life Expectancy Trend Line: Shows weighted life expectancy progression over time.
GDP Trend Area Chart: Shows growth patterns in total GDP over years.
World Choropleth Map: Displays country-wise GDP per capita distribution.
Top 10 GDP Countries Chart: Identifies strongest economies in selected year.

9. Implementation Notes

The app uses Streamlit's caching to improve performance for repeated runs. Weighted averages are used for life expectancy and GDP per capita to prevent small-population countries from disproportionately affecting global metrics. Plotly visualizations are fully interactive and adapt to filtered data.

10. Testing and Validation

- Verified that filter selections update all metrics and charts consistently.
- Validated weighted calculations against expected numeric behavior.
- Checked CSV export correctness for selected columns and year.
- Confirmed app loads in wide-layout and renders on standard desktop browser.

11. Outcomes and Benefits

This project demonstrates practical data analytics skills by converting raw historical data into actionable visual insights. It is suitable for academic presentation because it combines data engineering, analytics, and storytelling in a single deployable application.

12. Limitations

- Dataset covers years only up to 2007.
- No predictive model is currently integrated.
- Insights depend on quality and scope of the source dataset.

13. Future Enhancements

- Integrate recent data from live APIs or official open datasets.
- Add machine learning forecasting for GDP and life expectancy.
- Introduce country clustering and anomaly detection.
- Deploy on Streamlit Cloud with role-based or public access options.
- Add report automation (PDF/Word generation from selected filters).

14. Conclusion

The Global Development Analytics Dashboard is a complete and meaningful final-year analytics project. It effectively demonstrates the ability to collect, transform, analyze, and present real-world data through an interactive interface. The project is technically sound, presentation-ready, and extensible for future work.

15. Project Files

- app.py - Main Streamlit dashboard application
- requirements.txt - Python dependencies
- README.md - Project overview and execution steps